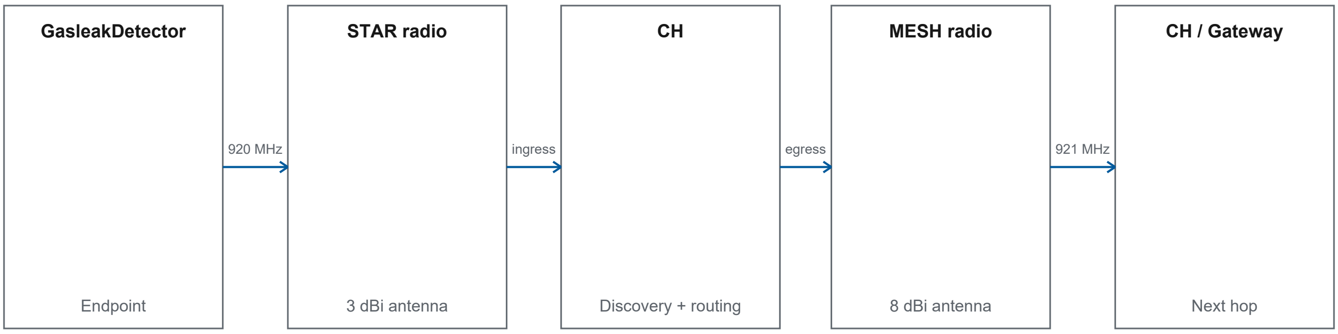
CH

Technical Datasheet Revision 4.0

Dual-radio node for LoRa STAR aggregation, dynamic routing, and multi-hop LoRa MESH forwarding.



PCB VARIANTS	RADIOS	ENERGY	ROUTING
Rectangle / Circle	2 x E22-900MM22S	Battery + 2 panels	Dynamic multi-hop

Key features

- Receives GasleakDetector traffic through Radio A/STAR.
- Selects a parent based on discovery, root reachability, and health.
- Forwards uplink and downlink through Radio B/MESH.
- Supports additional transit CH nodes when a direct Gateway route is not selected.

Technical overview

CH bridges the LoRa STAR endpoint domain and LoRa MESH backbone without permanently fixing a parent.

Functional profile

Ingress	LoRa STAR
Egress	LoRa MESH
Radios	2
Variants	2 PCB forms

Operational interfaces

Serving CH	Receives STAR frames from its served GasleakDetector.
Transit CH	Forwards MESH frames without becoming the endpoint data source.
Gateway root	Final route destination; not a static parent for every CH.

Contents

1 Product identification . . . 3

1.1 Abbreviations . . . 3

2 Overview and hardware . . . 3

2.1 Product overview . . . 3

2.2 CH role in the network . . . 3

2.3 PCB variants and firmware compatibility . . . 3

2.4 Primary hardware components . . . 4

2.5 Dual-radio architecture . . . 4

3 Radio interfaces . . . 4

3.1 LoRa STAR characteristics . . . 4

3.2 LoRa MESH characteristics . . . 5

3.3 Defaults, NVS, and device readback . . . 5

3.4 Nominal EIRP calculation . . . 5

4 Routing and transport . . . 5

4.1 Discovery and parent candidates . . . 5

4.2 Active parent, alternate, and loop prevention . . . 6

4.3 Multi-hop forwarding . . . 6

4.4 Normal telemetry: cache and server pull . . . 6

4.5 Alarm push and return-to-clear . . . 6

4.6 Parent failover . . . 7

5 Power system . . . 7

5.1 CH energy architecture . . . 7

5.2 CH battery configuration . . . 7

5.3 Solar panels and charging . . . 7

5.4 Battery telemetry and protection boundary . . . 8

6 Routing and operational capacities . . . 8

6.1 Firmware resource capacities . . . 8

6.2 CH operating states . . . 8

6.3 Parent-selection constraints . . . 8

6.4 Routing and transport retention . . . 9

7 Commissioning and references . . . 9

7.1 Commissioning and Operator Hub . . . 9

7.2 References and revision history . . . 9

1 Product identification

Product	CH
Product status	Engineering Prototype
Document revision	4.0
Technical baseline	Firmware 0.8.0 | Protocol 0.2.0
Issuer	Lab IoT ITB

1.1 Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
CH	Communication Hub / field communication node	RF	Radio frequency
EIRP	Equivalent Isotropically Radiated Power	STAR	GasleakDetector-to-CH radio domain
MESH	Backbone radio domain between CH and Gateway	TX	Transmit
NVM/NVS	Non-volatile configuration storage	VBAT	Battery voltage read by the device

2 Overview and hardware

2.1 Product overview

CH bridges the LoRa STAR endpoint domain and LoRa MESH backbone without permanently fixing a parent.

Ingress	LoRa STAR	**Egress**	LoRa MESH
Radios	2	**Variants**	2 PCB forms

- Receives GasleakDetector traffic through Radio A/STAR.
- Selects a parent based on discovery, root reachability, and health.
- Forwards uplink and downlink through Radio B/MESH.
- Supports additional transit CH nodes when a direct Gateway route is not selected.

2.2 CH role in the network

A CH can act as the serving CH for local endpoints and as a transit CH for other nodes.

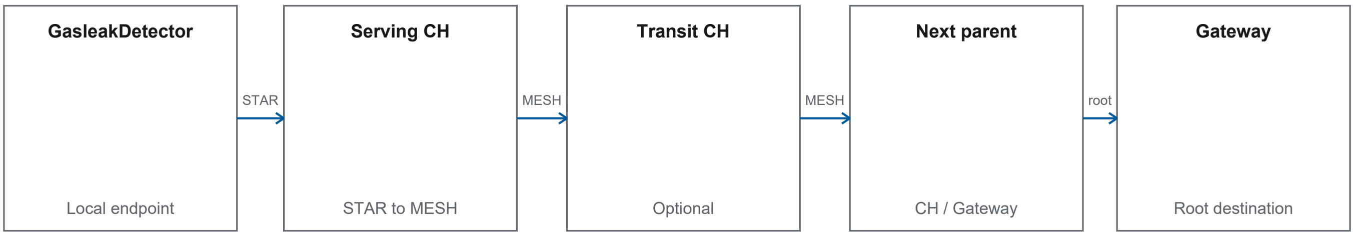


Figure 1. Serving and transit roles in the field path.

Serving CH	Receives STAR frames from its served GasleakDetector.
Transit CH	Forwards MESH frames without becoming the endpoint data source.
Gateway root	Final route destination; not a static parent for every CH.

2.3 PCB variants and firmware compatibility

Rectangle/small and Circle/large use different hardware mappings, so the firmware target must be selected explicitly.

Product variant	Design basis	Operator Hub selection	Description
CH Rectangle	Rectangle hardware profile	CH Rectangle	Rectangle-board-specific mapping
CH Circle	Circle hardware profile	CH Circle	Circle-board-specific mapping

Stage	Description
Select board form	Rectangle or Circle
Select firmware variant	Target must match the hardware profile
Upload	Operator Hub rejects hardware fallback
Readback	Version, identity, and board variant are checked

Figure 2. Board and firmware compatibility gate.

> Note: The two firmware variants are not interchangeable without selecting the correct hardware target.

2.4 Primary hardware components

Both boards share the same functional architecture even though layout and pin mapping differ.

Function	Design component	Role
MCU	ESP32-S3-WROOM-1U-N16R8	Runtime, routing, configuration, and status
Radio A	EBYTE E22-900MM22S	LoRa STAR domain
Radio B	EBYTE E22-900MM22S	LoRa MESH domain
Charger	TI BQ25185DLHR	PCB charging/solar interface
Watchdog	TI TPL5010DDCR	Hardware watchdog/timing

> Note: Component temperature, safety, and compliance ratings do not become CH product ratings.

2.5 Dual-radio architecture

STAR and MESH domains are physically and logically separated with different radios and defaults.

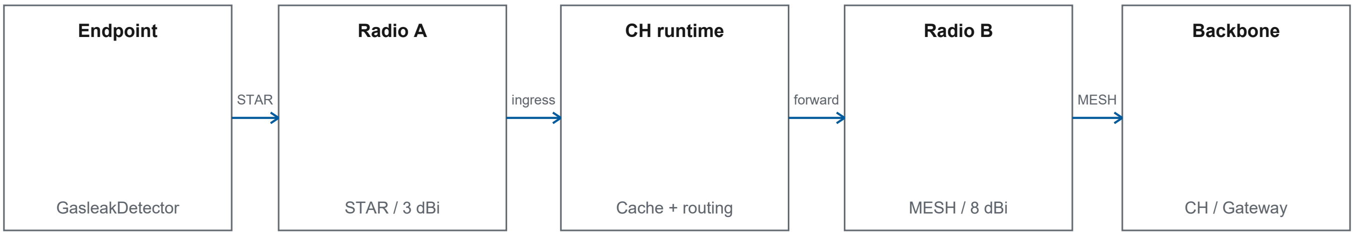


Figure 3. Separation of the two CH radio domains.

- Radio A and Radio B use the SX1262 driver.
- Normal operation requires successful STAR and MESH initialization.
- STAR and MESH use separate antennas for their respective radio domains.

3 Radio interfaces

3.1 LoRa STAR characteristics

STAR is the local link from GasleakDetector to its serving CH.

Parameter	Default	Status
Carrier	920.0 MHz	Configuration accepted from 920-923 MHz
Bandwidth	125 kHz	Firmware default
Spreading factor	SF7	Firmware default
Coding rate	4/5	Firmware default
TX setting	17 dBm	Setting, not measured output
CH antenna	3 dBi	Product configuration

> Note: Actual configuration may come from NVS. Device readback, not the default table, is the deployment reference.

3.2 LoRa MESH characteristics

MESH is the CH-to-CH and CH-to-Gateway backbone.

Parameter	Default	Status
Carrier	921.0 MHz	Configuration accepted from 920-923 MHz
Bandwidth	125 kHz	Firmware default
Spreading factor	SF9	Firmware default
Coding rate	4/5	Firmware default
TX setting	17 dBm	Setting, not measured output
CH antenna	8 dBi	Product configuration

3.3 Defaults, NVS, and device readback

Firmware distinguishes defaults, stored values, and actual values read back after a write.

Stage	Description
Build default	STAR 920.0 MHz; MESH 921.0 MHz
Input validation	Carrier must be within 920-923 MHz inclusive
Storage	STAR/MESH configuration is stored in NVS
Readback	Write is verified before success ACK
Next boot	Valid configuration is reloaded

Figure 4. Controlled RF-configuration cycle.

> Note: Engineers must record unit readback after commissioning.

3.4 Nominal EIRP calculation

The values below are arithmetic TX setting plus antenna gain before losses; they are not RF measurements.

Domain	TX setting	Antenna gain	Nominal before loss
STAR	17 dBm	3 dBi	20 dBm
MESH	17 dBm	8 dBi	25 dBm

- Cable/connector loss is not included in the nominal calculation.

> Note: If regulatory documentation requires EIRP, use accredited-laboratory results and the final antenna configuration.

4 Routing and transport

4.1 Discovery and parent candidates

CH builds parent candidates through discovery and does not use a fixed parent identity.

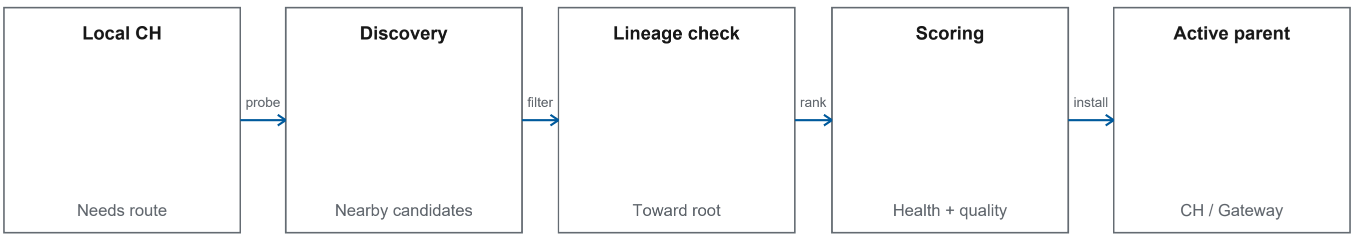


Figure 5. Active-parent selection flow.

- A candidate must have lineage that reaches the Gateway root.
- Loops and invalid lineage are rejected.
- Parent and alternate can be stored after stabilization.

4.2 Active parent, alternate, and loop prevention

Routing maintains a path toward the Gateway while preventing the parent chain from looping back to the origin node.

Control	Function	Outcome
Root reachability	Ensures candidate has a path to Gateway	Candidate without root is rejected
Lineage guard	Checks parent chain	Looping candidate is rejected
Parent health	Monitors active-parent response	Can trigger replacement
Alternate parent	Keeps a backup candidate	Can be used before rediscovery

> Note: The actual route depends on field discovery; the datasheet does not name any parent as a permanent pair.

4.3 Multi-hop forwarding

A frame can pass through zero or more transit CH nodes before reaching the Gateway.

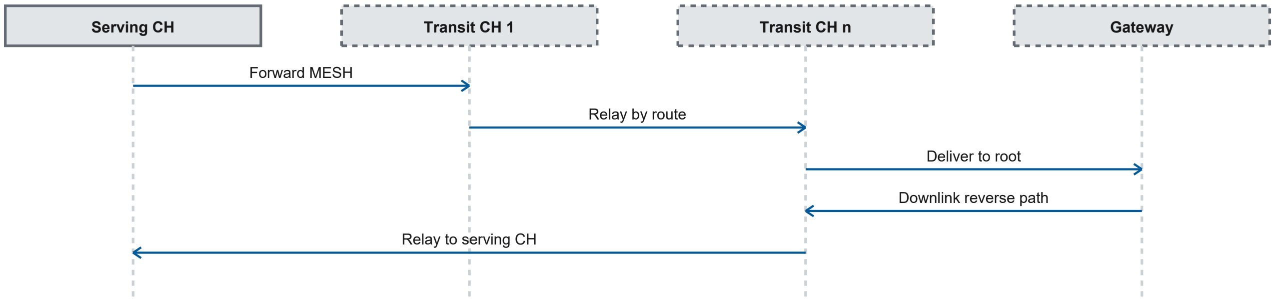


Figure 6. Multi-hop uplink and downlink forwarding.

4.4 Normal telemetry: cache and server pull

Normal telemetry does not use the same push pattern as alarms; data is cached at CH and sent in response to a pull.

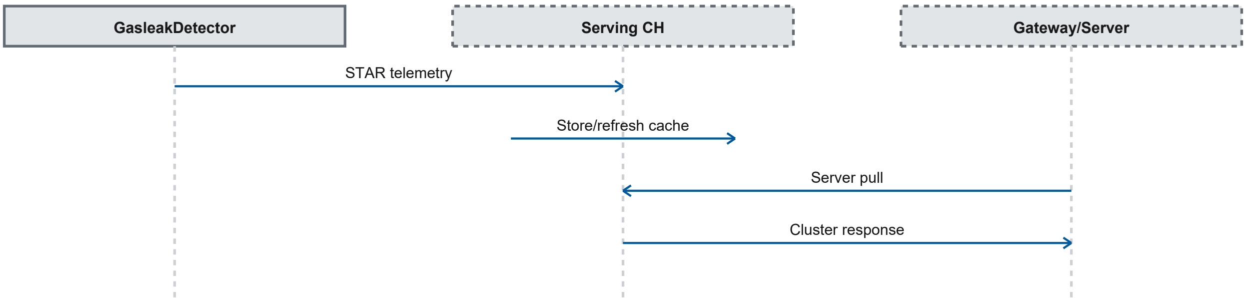


Figure 7. Cache/pull path for normal telemetry.

- Cache separates endpoint reception from the server request cycle.
- Delivery to the Server occurs when a pull request reaches the serving CH.

4.5 Alarm push and return-to-clear

Alarm and return-to-clear events use a push path so they do not wait for the normal telemetry cycle.

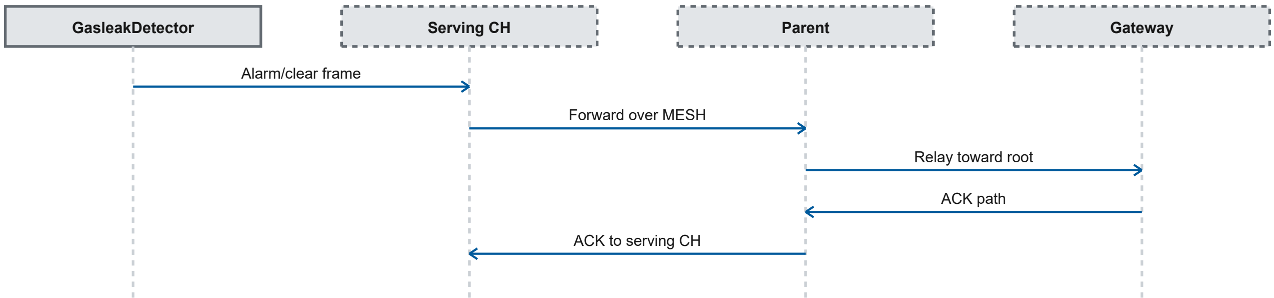


Figure 8. Push path for alarm and return-to-clear.

4.6 Parent failover

Failover can use an alternate parent or return to discovery when active-parent health degrades.

Stage	Description
Active parent	HELLO/ACK and health are monitored
Failure trigger	Timeout, HELLO ACK failure, or alarm ACK failure
Alternate	Used when still valid
Rediscovery	Finds new candidates when needed
New route	Parent is determined by discovery

Figure 9. Parent-failover state flow.

> Note: Multi-hop and failover have been validated at the ITB IoT Lab. The replacement parent is selected dynamically through discovery.

5 Power system

5.1 CH energy architecture

The product configuration combines parallel batteries and two parallel solar panels through on-board charging hardware.

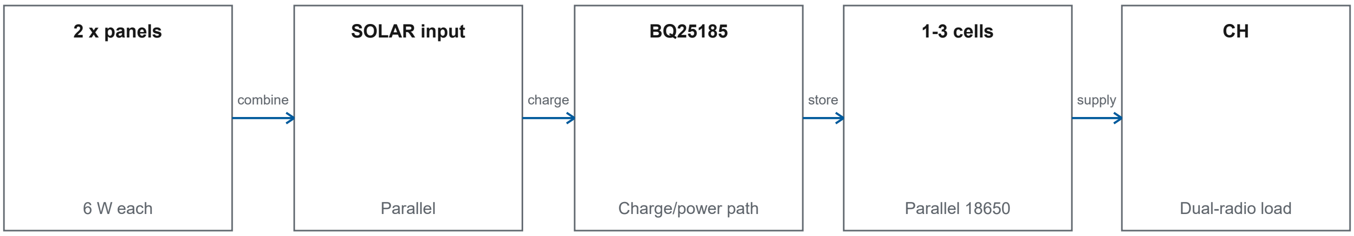


Figure 10. CH energy topology.

> Note: The product configuration uses 1-3 parallel 18650 cells and two parallel 6 W panels through the on-board charging path.

5.2 CH battery configuration

CH uses one to three LiitoKala 18650 cells connected in parallel.

Parameter	Configuration	Note
Type	LiitoKala Lii-King4000 18650	Cell-label identification
Quantity	1-3	Deployment-dependent
Capacity label	4000 mAh / cell	Not a usable-capacity test
Topology	Parallel	Maintains the single-cell voltage domain

> Note: The 4000 mAh value is the nameplate capacity of each cell.

5.3 Solar panels and charging

Two 6 W panels are connected in parallel to the CH solar interface.

Panel count	2	**Nameplate**	6 W each
Topology	Parallel	**Combined**	12 W nameplate

Board interface	One SOLAR input into the power/charge path.
Controller	TI BQ25185DLHR

> Note: 12 W is the sum of two panel nameplates, not measured field output.

5.4 Battery telemetry and protection boundary

Firmware reads and reports battery voltage but does not use low voltage to block product operation.

Stage	Description
VBAT sensing	Voltage is read by CH
Status	Value is reported read-only
Boot/TX	Not blocked by VBAT threshold
Low-power	Not automatically triggered by undervoltage

Figure 11. VBAT read-only function boundary.

> Note: VBAT is used for diagnostic telemetry; firmware does not disconnect power or enter low-power mode automatically from that reading.

6 Routing and operational capacities

6.1 Firmware resource capacities

The following table states the number of entries that can be retained concurrently by the CH routing and transport services.

Resource	Capacity	Use
Parent candidates	8	Discovery candidates evaluated for parent selection.
Node cache	32	Information about nodes known to the CH.
Alarm queue	8	Alarms awaiting transport processing.
Transmission queue	8	Items awaiting radio transmission.
Pending downlink	16	Downlink commands or responses awaiting completion.

6.2 CH operating states

The state machine separates initialization, route establishment, connected operation, failover, and recovery.

State	Function	Product profile
Startup	Starts services and base configuration.	Active
Battery-wait state	Entered briefly at boot; the read-only profile bypasses the readiness threshold.	Proceeds directly to radio initialization
Radio initialization	Initializes the STAR and MESH radios.	Active
Network discovery	Performs discovery and selects a parent.	Active
Connected	Forwards traffic using the selected route.	Active
Low-power state	State available for low-power operation.	Not triggered automatically by the battery reading
Parent failover	Searches for a replacement when the parent is no longer eligible.	Active
Recovery	Restores radio and routing services after a disturbance.	Active

> Note: The product profile reads battery voltage for telemetry. The value is not used to prevent boot, stop transmission, or activate the low-power state automatically.

6.3 Parent-selection constraints

Parent selection is dynamic and discovery-driven, with stability constraints that prevent excessively rapid route changes.

Parameter	Value	Operational effect
Candidate capacity	8	Limits the number of candidates evaluated at one time.
Parent health timeout	16 min	A parent not seen within this interval triggers failover evaluation in the normal product profile.
Minimum parent dwell	5 min	Prevents routine switching before the minimum time has elapsed.
Switch margin	15 dB	A replacement candidate must provide sufficient improvement for a routine switch.
Stable discoveries	4	Applies to routine background switching and parent/alternate persistence; it is not an initial-join or failover gate.

> Note: The replacement parent is not static; failover selects an eligible discovery candidate.

6.4 Routing and transport retention

Routing and downlink data have bounded retention periods so that old information is not used indefinitely.

Data	Retention	Entry limit
Parent without refresh	16 min	1 active parent
Pending downlink	Default 30 min; command TTL overrides	16
Node cache	60 min	32

- Queues and caches have fixed capacities; the number of retained entries does not exceed the tabulated limits.
- Failover uses the available discovery results, not a programmed fixed replacement-parent list.
- The STAR and MESH radios are handled as separate interfaces so that node reception and MESH forwarding can be managed independently.

7 Commissioning and references

7.1 Commissioning and Operator Hub

Commissioning locks board form, firmware variant, identity, and configuration readback before unit acceptance.

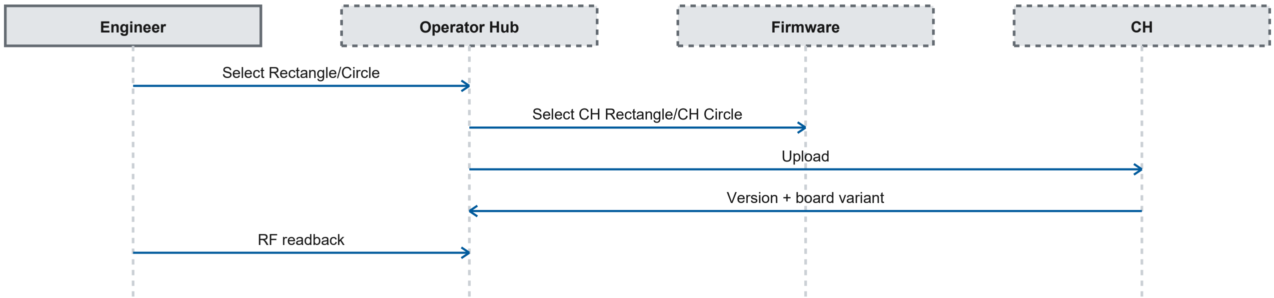


Figure 12. CH commissioning gates.

Interface	Scope
Simple Operator Hub	Board/firmware selection and simplified configuration controls.
CH Expert Operator	More complete STAR and MESH configuration/readback.

7.2 References and revision history

Component references are used for identification, not to elevate assembled-product claims.

Reference	Document
R1	Espressif - ESP32-S3-WROOM-1/1U Datasheet v1.8
R2	EBYTE - E22-900MM22S module documentation
R3	Texas Instruments - BQ25185 product datasheet
R4	Texas Instruments - TPL5010 product datasheet

Rev	Change
3.0	Updated technical structure and specifications.